

# Relation Algebra

## §11. Relator

(11a)

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from  $R$  to  $S$  IS  $R \sqsubseteq S$ . A **relator**  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a 2-functor between allegories:

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving  $\circ$  is **not** asked for —  $\circ$  is an identity-on-objects involution  $\mathcal{C}^{\circ} \rightarrow \mathcal{C}$ , no part of the 2-category.

(11b)

**the statement**

$\text{Map}(f) \implies \text{Map}(F(f))$  and  $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$ .

Over a **tabular** allegory a functor is a relator  $\iff$  it preserves  $\circ$ .

$F(R^\circ) = F(R)^\circ$  for every  $R$ , so  $F(R)^\circ$  needs no bracket.

Two relators agreeing on maps are equal.

$\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$

$F(R \cap S) \sqsubseteq F(R) \cap F(S)$ , and strictly.

$\text{PreservesRecip}(F) \implies F(\text{dom}(R)) = \text{dom}(F(R))$

The **power relator**  $P \multimap x \sqsubseteq P(R) \ y \iff (\forall a \in x. \exists b \in y. a R b) \wedge (\forall b \in y. \exists a \in x. a R b)$  — is where the fourth is strict: for  $R = \{(a_1, b_1), (a_2, b_2)\}$  and  $S = \{(a_1, b_2), (a_2, b_1)\}$  the pair  $(\{a_1, a_2\}, \{b_1, b_2\})$  is in  $P(R) \cap P(S)$ , while  $R \cap S = \emptyset$ .

### §11.1. Fork $\langle R, S \rangle$

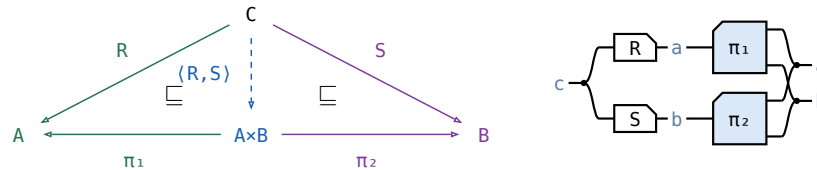
(11.1a)

The **fork** of  $R : \mathcal{C} \rightarrow \mathcal{A}$  and  $S : \mathcal{C} \rightarrow \mathcal{B}$  is  $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$ , where  $(\pi_1, \pi_2)$  is the tabulation of  $\tau$ .

(11.1b)

$$\langle R, S \rangle \pi_1 = \text{dom}(S) R \quad \langle R, S \rangle \pi_2 = \text{dom}(R) S$$

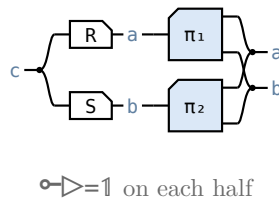
(11.1c)



A domain is coreflexive, so  $\langle R, S \rangle \pi_1 \sqsubseteq R$ , with equality exactly when  $S$  is entire; for maps both triangles commute and  $\langle f, g \rangle$  is unique. In  $\mathbf{Rel}$ ,  $c \sqsubseteq \langle R, S \rangle (a, b)$  iff  $c R a$  and  $c S b$  — copy  $c$ , then  $R$  on one strand and  $S$  on the other, which is  $\triangleleft (R \otimes S)$  on the right.

No  $\circ$  survives the translation.  $\pi_1 = 1 \otimes \circ$  discards the second component, so  $\pi_1^\circ = 1 \otimes \circ$  **creates** one out of nothing, and  $\cap$  is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

(11.1d)

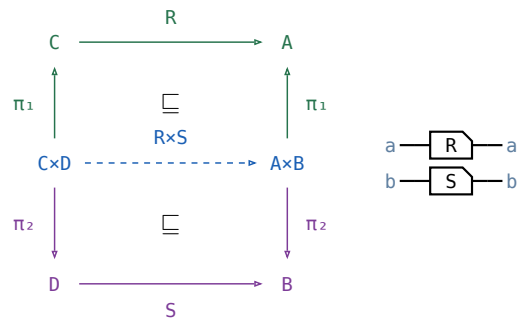


#### §11.1.1. Relational product $R \times S$

(11.1.1a)

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$ , a relator in each argument but no longer a categorical product.

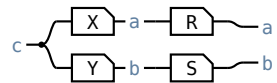
(11.1.1b)



Right-then-up is  $(R \times S)\pi_1$ , up-then-right is  $\pi_1 R$ , and  $(R \times S)\pi_1 \sqsubseteq \pi_1 R$ , equality when  $S$  is entire. In  $\mathbf{Rel}$ ,  $(c, d) (R \times S) (a, b)$  iff  $c R a$  and  $d S b$  — two strands side by side, no copy dot:  $R \times S = R \otimes S$ .

### §11.1.2. Absorption

For  $X : E \rightarrow C$  and  $Y : E \rightarrow D$ ,  $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$ . Both sides are this picture:



(11.1.2a)

(11.1.2b)

| the law                                                                                                                                                                                                                                                                                                                                                            | picture |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1 $R \times S \triangleq (\pi_1 R, \pi_2 S)$                                                                                                                                                                                                                                                                                                                       |         |
| 2 $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$<br>both sides are the same strokes — absorption. Its $\sqsubseteq$ half is that half at $S := \mathbb{1}$ and at $R := \mathbb{1}$ — stages of their proof of this row; the corollary is the $(R \times S) (U \times V) = (RU) \times (SV)$ it yields, at $R := \mathbb{1}$ and $V := \mathbb{1}$ . |         |
| 3 $\langle R, S \rangle \pi_1 = \text{dom}(S) R$<br>(11.1b)                                                                                                                                                                                                                                                                                                        |         |
| 4 $\langle R, S \rangle \pi_2 = \text{dom}(R) S$<br>(11.1b)                                                                                                                                                                                                                                                                                                        |         |
| 5 $\langle X, Y \rangle (R, S)^\circ = XR^\circ \cap YS^\circ$                                                                                                                                                                                                                                                                                                     |         |
| 6 $\langle R, S \rangle^\circ (U, V) \sqsubseteq (R^\circ U) \times (S^\circ V)$                                                                                                                                                                                                                                                                                   |         |
| 7 $\text{Map}(f) \implies f \langle R, S \rangle = \langle fR, fS \rangle$<br>$f$ a map; it fails for an arbitrary arrow;                                                                                                                                                                                                                                          |         |
| 8 $F(R \times S) \text{unzip}(F) \sqsubseteq \text{unzip}(F) (F(R) \times F(S))$<br>$\text{unzip}(F) \triangleq (F(\pi_1), F(\pi_2))$ ;<br>only $\sqsubseteq$ for an arbitrary $R, S$                                                                                                                                                                              |         |
| 9 $g = \text{curry}(f) \iff (g \times \mathbb{1}) \text{eval} = f$<br>reading $\times$ as the relational product, does $\mathbf{Rel}$ have exponentials?                                                                                                                                                                                                           |         |

## §11.2. Coproduct $[R, S] : A + B \rightarrow C$

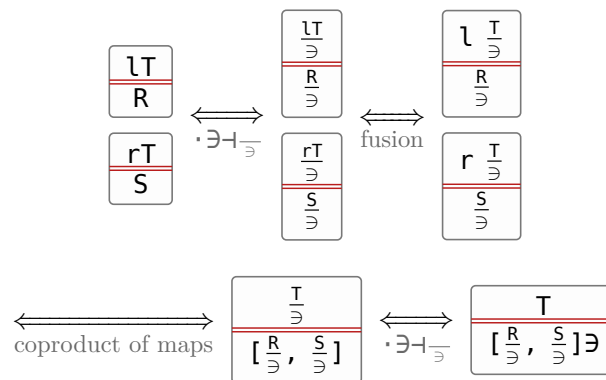
(11.2a)

| the statement                                                                                                                                                                                                    | picture |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $[R, S] \triangleq \mathfrak{l} \circ R \cup r \circ S$<br>The tape is the union — a particle entering at <b>A+B</b> takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint. |         |
| $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$                                                                                                                                                        |         |
| $R + S \triangleq [R\mathfrak{l}, S\mathfrak{r}]$                                                                                                                                                                |         |
| $\mathfrak{l}[R, S] = R, r[R, S] = S$<br>$\mathfrak{l}T = R \wedge rT = S \implies T = [R, S]$                                                                                                                   |         |
| $\mathfrak{l}\mathfrak{l}^\circ = 1, r\mathfrak{r}^\circ = 1$                                                                                                                                                    |         |
| $\mathfrak{l}r^\circ = \emptyset, r\mathfrak{l}^\circ = \emptyset$                                                                                                                                               |         |
| $\mathfrak{l}^\circ \mathfrak{l} \cup r^\circ r = 1$                                                                                                                                                             |         |
| $[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$                                                                                                                                                                 |         |

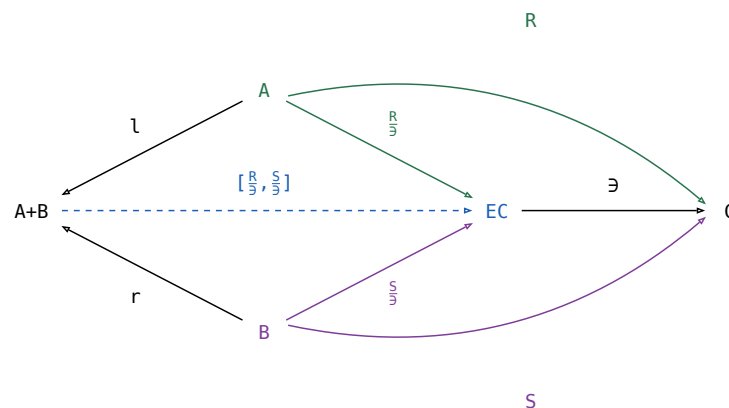
### §11.2.1. $[R, S] \triangleq [\frac{R}{\exists}, \frac{S}{\exists}] \exists$

The universal-property row is not free:  $\mathfrak{l}, r$  were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because  $\frac{\exists}{\exists}$  sends an arrow  $A \rightarrow C$  to a map  $A \rightarrow EC$  reversibly, so the map coproduct can be applied underneath it. For any  $T : A+B \rightarrow C$ ,

(11.2.1a)



(11.2.1b)



(11.2a)

Nothing here holds only up to  $\Xi$ : every triangle commutes on the nose, which is the difference from the fork above. The border spells  $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$ , and pushing  $\exists$  into the union that  $[\cdot, \cdot]$  on maps already is turns that back into the definition,  $(\frac{R}{\exists} \cup \frac{S}{\exists}) \exists = \frac{R}{\exists} \exists \cup \frac{S}{\exists} \exists = l \circ R \cup r \circ S$ .

(11.2.1c)

| the law                                                                                                                                                                                                                       | picture |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $[R, S] \triangleq l \circ R \cup r \circ S$<br>(11.2a)'s first row                                                                                                                                                           |         |
| $R + S \triangleq [Rl, Sr]$                                                                                                                                                                                                   |         |
| $[U, V] \circ [R, S] = U \circ R \cup V \circ S$<br>(11.2a)'s last row;                                                                                                                                                       |         |
| $[1, 0] = l \circ, [0, 1] = r \circ$<br>$[l, r] = 1$<br>which is (5.9)                                                                                                                                                        |         |
| prove (5.11), and say why duality<br>does not carry it over from the<br>product law                                                                                                                                           |         |
| $(R + S) \cap [P, Q] [U, V] \circ =$<br>$(R \cap P U \circ) + (S \cap Q V \circ)$<br>$[P, Q] [U, V] \circ$ is a full $2 \times 2$ of<br>composites; $R + S$ is diagonal, so the<br>meet cuts the two off-diagonal<br>branches |         |

### §11.3. The power relator $P(R)$

(11.3a)

For  $R : A \rightarrow B$ ,

$$P(R) \triangleq (\epsilon \setminus R \epsilon) \cap (\exists R / \exists)$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R) / \exists) \cap (\exists / (\exists R))^{\circ}$$

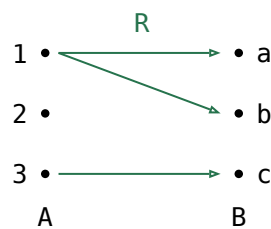
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of  $xs$  is related by  $R$  to some element of  $ys$ , and conversely.

(11.3b)

|                                           | in Rel                                                                                              | in words                                                                        | $\text{im}(xs)$              | partner                                       |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------|-----------------------------------------------|
| $(\exists R) / \exists$                   | $\forall y \in ys. \exists x \in xs. x \ R \ y$                                                     | $\forall y. \text{ some } xs \ R \ y$                                           | $ys \subseteq \text{im}(xs)$ | every $y$ has a partner                       |
| $(\exists / (\exists R))^{\circ}$         | $\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$                                     | every $y$ with some $xs \ R \ y$ is in $ys$                                     | $ys \supseteq \text{im}(xs)$ | $ys$ leaves out no partner                    |
| $E(R)$                                    | $ys = \{y \mid \exists x \in xs. x \ R \ y\}$                                                       | $ys = \text{ every } y \text{ with some } xs \ R \ y$                           | $ys = \text{im}(xs)$         | every $y$ has a partner, and none is left out |
| $((\exists R^{\circ}) / \exists)^{\circ}$ | $\forall x \in xs. \exists y \in ys. x \ R \ y$                                                     | $\forall x. x \ R \ \text{ some } ys$                                           | —                            | every $x$ has a partner                       |
| $P(R)$                                    | $\forall y \in ys. \exists x \in xs. x \ R \ y$ and $\forall x \in xs. \exists y \in ys. x \ R \ y$ | $\forall y. \text{ some } xs \ R \ y$ and $\forall x. x \ R \ \text{ some } ys$ | —                            | every $x$ and every $y$ has a partner         |

(11.3c)



| $xs$          | $(\exists R) / \exists$ | $(\exists / (\exists R))^{\circ}$ | $E(R)$      | $((\exists R^{\circ}) / \exists)^{\circ}$ | $P(R)$        |
|---------------|-------------------------|-----------------------------------|-------------|-------------------------------------------|---------------|
| $\emptyset$   | $\emptyset$             | all 8 subsets of $abc$            | $\emptyset$ | all 8 subsets of $abc$                    | $\emptyset$   |
| $\{1\}$       | $\emptyset, a, b, ab$   | $ab, abc$                         | $ab$        | $a, b, ab, ac, bc, abc$                   | $a, b, ab$    |
| $\{2\}$       | $\emptyset$             | all 8 subsets of $abc$            | $\emptyset$ | none                                      | none          |
| $\{3\}$       | $\emptyset, c$          | $c, ac, bc, abc$                  | $c$         | $c, ac, bc, abc$                          | $c$           |
| $\{1, 2\}$    | $\emptyset, a, b, ab$   | $ab, abc$                         | $ab$        | none                                      | none          |
| $\{1, 3\}$    | all 8 subsets of $abc$  | $abc$                             | $abc$       | $ac, bc, abc$                             | $ac, bc, abc$ |
| $\{2, 3\}$    | $\emptyset, c$          | $c, ac, bc, abc$                  | $c$         | none                                      | none          |
| $\{1, 2, 3\}$ | all 8 subsets of $abc$  | $abc$                             | $abc$       | none                                      | none          |

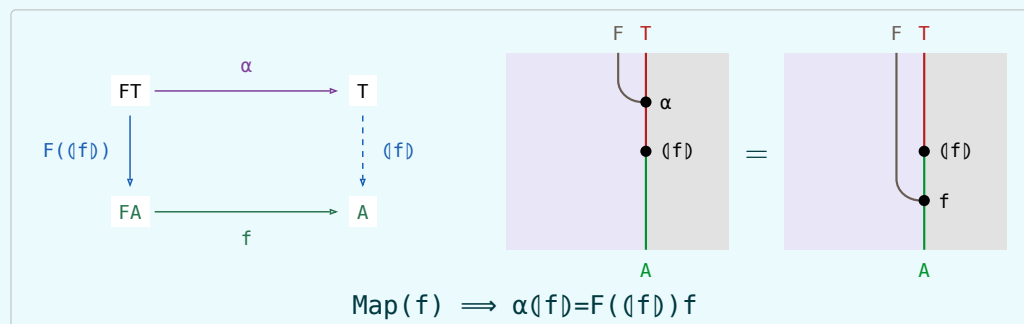
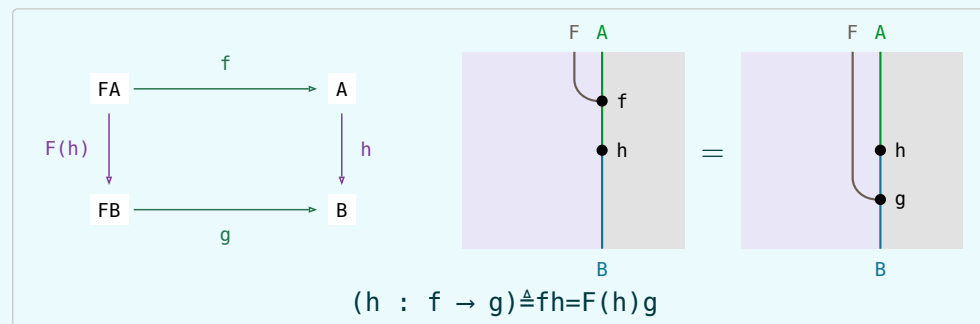
(11.3d)

| law                                                                                             | the reading                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $X \sqsubseteq P(R) \iff X \sqsubseteq \exists R$ and $X^{\circ} \sqsubseteq \exists R^{\circ}$ | One containment, and the same one at $R^{\circ}$ — which is the definition read off the two divisions. Hence $P(R^{\circ}) = P(R)^{\circ}$ , and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$ .                                                                                                              |
| $P(1) = 1$                                                                                      | The straightness axiom verbatim: extensionality is $P$ 's unit law.                                                                                                                                                                                                                                              |
| $\text{Map}(f) \implies P(f) = E(f)$                                                            | In Rel, $xs \ P(f) \ ys \iff ys = \{f(a) \mid a \in xs\}$ . The half at $f^{\circ}$ says every $a \in xs$ has its $f(a)$ on $ys$ ; $f$ has just the one image per $a$ , so that already says $ys$ contains everything $xs$ reaches, which is the fraction's second half. For a map the two definitions coincide. |
| $P(RS) = P(R)P(S)$                                                                              | $\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$ .                                                                                                                                                      |

## §11.4. Initial algebra

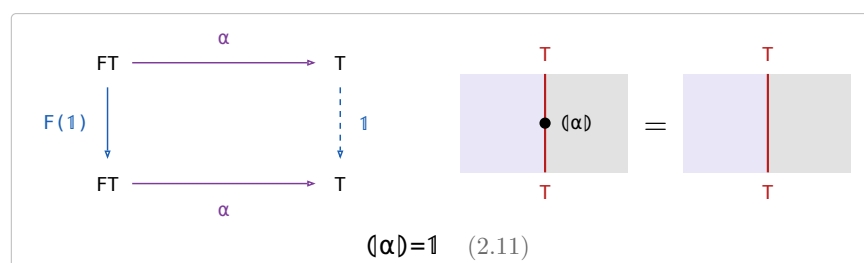
(11.4a)

An **F-algebra** is a map  $f : F(A) \rightarrow A$ ;  $A$  is its **carrier**. An **F-homomorphism** from  $f : F(A) \rightarrow A$  to  $g : F(B) \rightarrow B$  is a map  $h : A \rightarrow B$  with  $f \circ h = F(h) \circ g$ . The **initial algebra**  $\alpha : F(T) \rightarrow T$  is the F-algebra with exactly one F-homomorphism  $\langle f \rangle : T \rightarrow A$  to every F-algebra  $f$ .



### §11.4.1. Reflection

(11.4.1a)

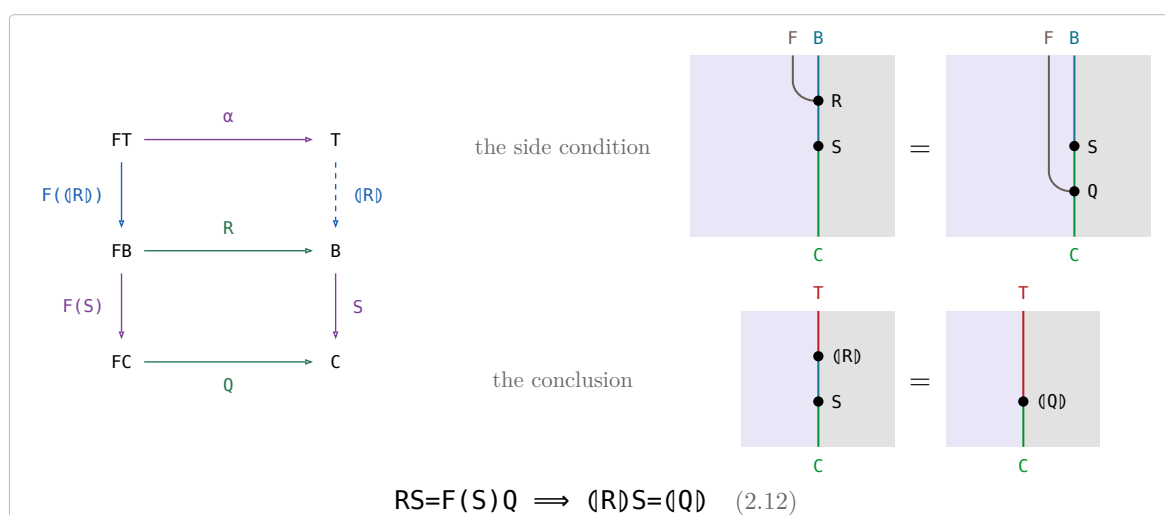


Taking a value apart with  $\alpha$  and putting it straight back is doing nothing.

### §11.4.2. Fusion: $(\langle R \rangle : FB) \text{ absorb } S : R \rightarrow Q \text{ (in } \text{Alg}(F) \text{ ) and becomes } \langle Q \rangle$

When  $S : B \rightarrow C$  is an F-homomorphism from  $R : F(B) \rightarrow B$  to  $Q : F(C) \rightarrow C$ , folding with  $R$  and then applying  $S$  is folding with  $Q$ .

(11.4.2a)



## §11.5. Type relator

(11.5a)

Let  $F$  be a binary relator with initial type  $(\alpha, T)$ , so  $T$  is a type functor.  $F(R, S)$  is its action on a pair, and  $F(X)$  abbreviates  $F(1, X)$ , the  $F$  of the reduce section. For every object  $A$  the initial algebra is  $\alpha : F(A, TA) \rightarrow TA$ , among the maps.  $T$  acts on an arrow  $R : A \rightarrow B$  by

$$T(R) = (F(R, 1) \alpha)$$

(11.5b)

| name                   | law                               | what it says                                                                                                                                                                                                |
|------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the defining equation  | $T(R) = (F(R, 1) \alpha)$         | Rebuild the structure with $\alpha$ , applying $R$ to the parameter on the way.                                                                                                                             |
| functor                | $T(1) = 1$ and $T(R)T(S) = T(RS)$ | Acting by the identity changes nothing, and two actions in a row are one action.                                                                                                                            |
| type functor fusion    | $T(f)(h) = (F(f, 1)h)$            | A relator action followed by a fold is a single fold — the intermediate structure is never built. The side condition holds because $F$ is a bifunctor — $F(R, 1)F(1, (Q)) = F(R, (Q)) = F(1, (Q))F(R, 1)$ . |
| naturality of $\alpha$ | $\alpha T(f) = F(f, T(f)) \alpha$ | Building and then mapping is the same as mapping the parts and then building, so $\alpha$ is natural from $G(R) = F(R, T(R))$ to $T$ .                                                                      |
| type relator           | $T(R)^\circ = T(R^\circ)$         | A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.                                                                                                        |

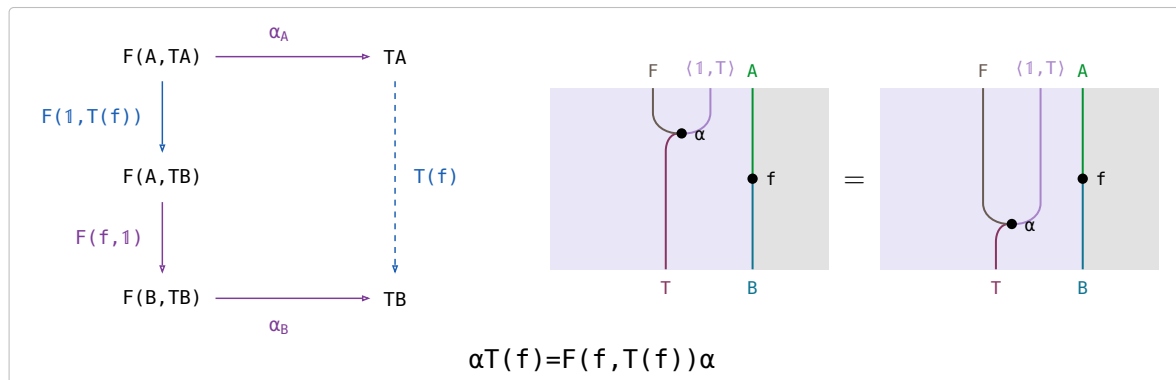
### §11.5.1. Type functor

(11.5.1a)

Let  $F$  be a bifunctor taking both the parameter  $A$  and the recursive position  $TA$ , with an initial algebra  $\alpha_A : F(A, TA) \rightarrow TA$  for every object  $A$ . Then  $T$  is a functor, acting on a map  $f : A \rightarrow B$  by

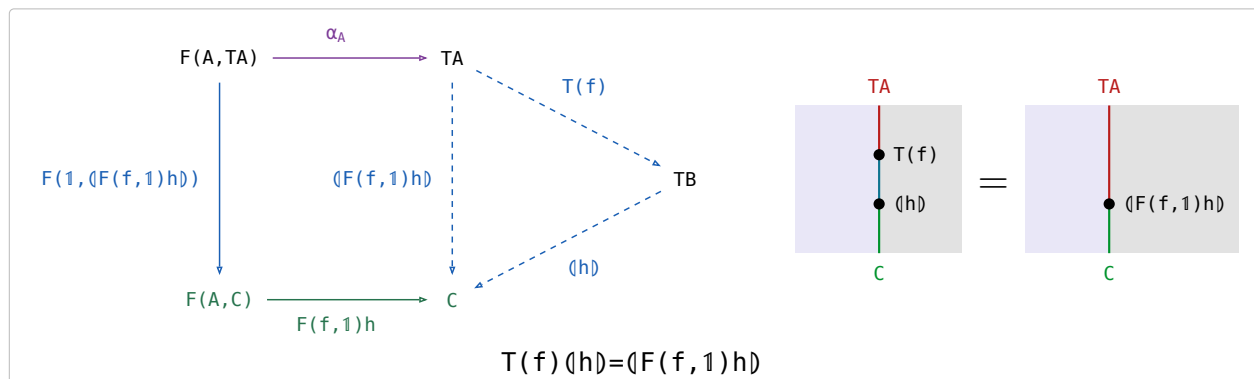
$$T(R) \triangleq (F(R, 1) \alpha)$$

(11.5.1b)



- $F : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$  is a bifunctor and a wire is a unary functor, so the two arguments are packed first:  $\langle 1, T \rangle : \mathcal{A} \rightarrow \mathcal{A} \times \mathcal{A}$  sends  $A$  to  $(A, TA)$ , and  $F(\langle 1, T \rangle(A))$  is  $F(A, TA)$ .
- The picture is three wires —  $F$ ,  $\langle 1, T \rangle$ , and the object — and the region between the first two is  $\mathcal{A} \times \mathcal{A}$ .
- $\alpha$  is then an ordinary natural transformation  $F \circ \langle 1, T \rangle \Rightarrow T$ : its bead eats the  $F$  and  $\langle 1, T \rangle$  wires, and the  $T$  wire is born under it.
- $F(f, T(f))$  costs no notation. It is the bead  $f$  on the object wire with  $\langle 1, T \rangle$  and  $F$  running past:  $\langle 1, T \rangle$  is what turns  $f$  into the pair  $(f, T(f))$ , and  $F$  is what applies it.
- The law is the naturality of  $\alpha$ , which is exactly the freedom to slide that  $f$  bead past it.

(11.5.1c)



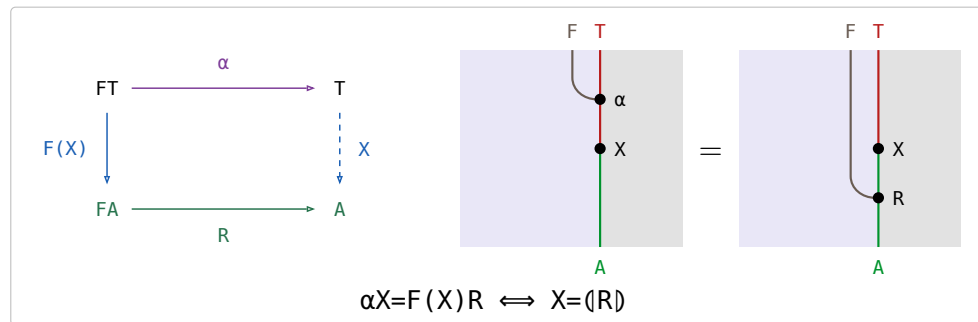
## §11.6. Reduce

(11.6a)

let  $F$  be a relator and has **initial algebra**  $\alpha : F(T) \rightarrow T$  in the subcategory of functions.  $\alpha$  is also initial in the allegory:

### §11.6.1. The defining equation

(11.6.1a)



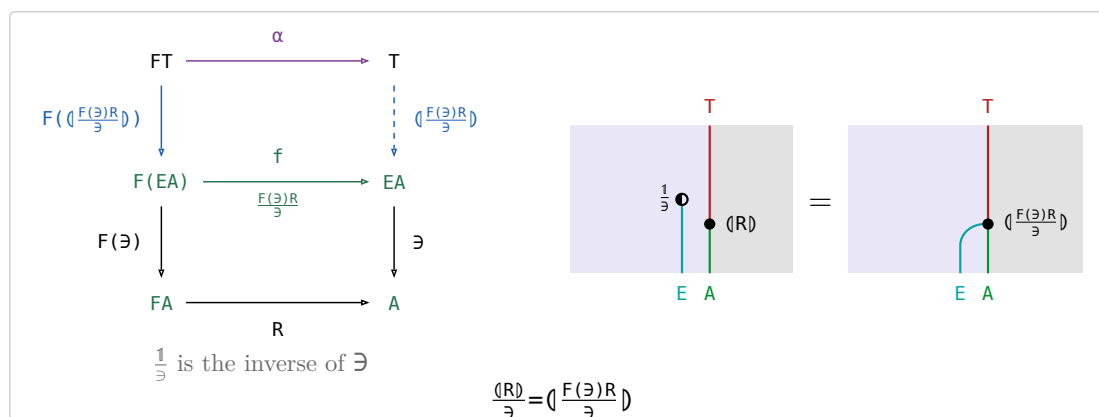
An  $F$ -algebra is a **weakened** natural transformation. A transformation  $F \Rightarrow 1$  on  $\mathcal{A}$  would need a component  $FX \rightarrow X$  at every object and a commuting square at every arrow, but  $F$ -Algebra only need it works on  $T$  and  $A$ .

(11.6.1b)

| $\llbracket [c, f] \rrbracket = \text{reduce}(c, f)$                                                                                                                     |                                                                                                                                        |                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| catamorphism: arrows of the form $\llbracket f \rrbracket$ are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing |                                                                                                                                        |                                                                                                                                             |
| part                                                                                                                                                                     | Nat                                                                                                                                    | [A]                                                                                                                                         |
| datatype                                                                                                                                                                 | $\text{Nat} ::= \text{zero} \mid \text{succ Nat}$                                                                                      | $[A] ::= \text{nil} \mid \text{cons}(A, [A])$                                                                                               |
| base functor $F$                                                                                                                                                         | $F(X) = 1 + X$                                                                                                                         | $F(X) = 1 + A \times X$                                                                                                                     |
| initial algebra $\alpha$                                                                                                                                                 | $\alpha = [\text{zero}, \text{succ}]$<br>$: 1 + \text{Nat} \rightarrow \text{Nat}$                                                     | $\alpha = [\text{nil}, \text{cons}]$<br>$: 1 + A \times [A] \rightarrow [A]$                                                                |
| the fold, pointwise                                                                                                                                                      | $\llbracket [c, f] \rrbracket(\text{zero}) = c$<br>$\llbracket [c, f] \rrbracket(\text{succ}(n)) = f(\llbracket [c, f] \rrbracket(n))$ | $\llbracket [c, f] \rrbracket(\text{nil}) = c$<br>$\llbracket [c, f] \rrbracket(\text{cons}(a, x)) = f(a, \llbracket [c, f] \rrbracket(x))$ |

### §11.6.2. $\llbracket (R) \rrbracket = \llbracket \frac{F(\exists)R}{\exists} \rrbracket \exists$

(11.6.2a)



(11.6.2b)

$$\begin{array}{c}
 \frac{\alpha X}{F(X)R} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \frac{\frac{\alpha X}{\exists}}{\frac{F(X)R}{\exists}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \frac{\frac{\alpha X}{\exists}}{\frac{F(\frac{X}{\exists})R}{\exists}} \\
 \xleftrightarrow{\text{relator, fusion twice}} \frac{\frac{\alpha \frac{X}{\exists}}{F(\frac{X}{\exists})} \frac{F(\exists)R}{\exists}}{\frac{F(\frac{X}{\exists})}{\exists}} \xleftrightarrow{\text{reduce of maps}} \frac{\frac{X}{\exists}}{\frac{(F(\exists)R)}{\exists}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \frac{X}{\frac{(F(\exists)R)}{\exists} \exists}
 \end{array}$$

the last two rows at  $X := (R)$ :

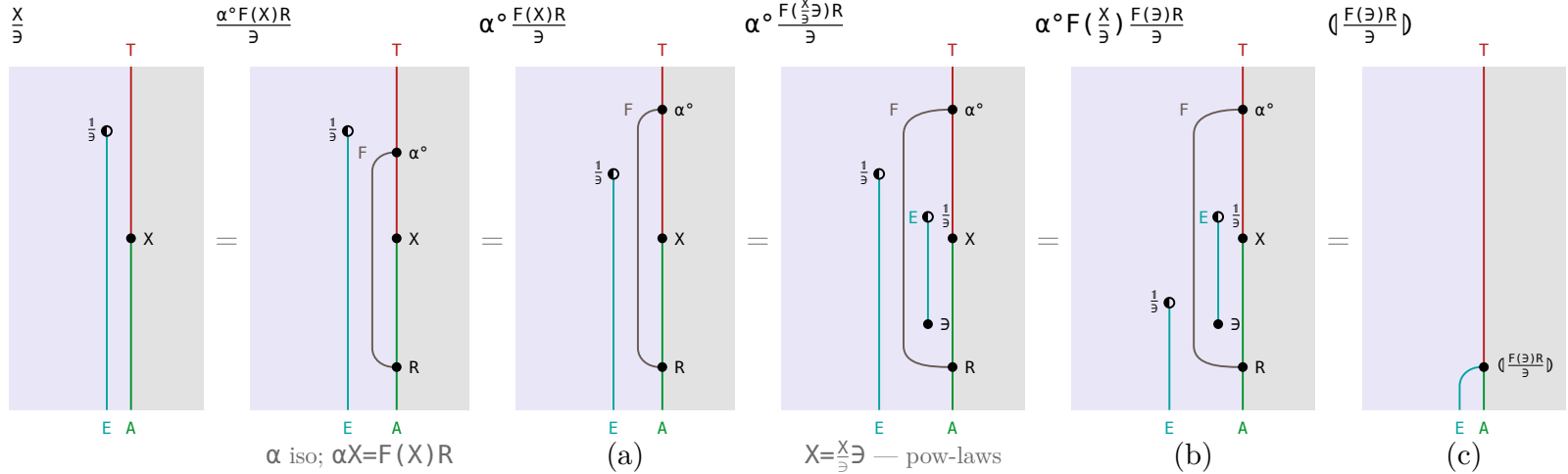
$$\alpha X = F(X)R \iff X = \langle R \rangle$$

(11.6.2c)

in a tabular allegory, with  $F$  a relator — so  $F(R^\circ) = F(R)^\circ$  — and  $\alpha : F(T) \rightarrow T$  its initial algebra, which every later fold theorem assumes: a relation  $X$  out of  $T$  satisfies the fold equation of the relation  $R$  exactly when it is  $\langle R \rangle$ , the fold of the map  $\frac{F(\exists)R}{\exists}$  followed by  $\exists$

$$\alpha X = F(X)R \implies \frac{X}{\exists} = \langle \frac{F(\exists)R}{\exists} \rangle$$

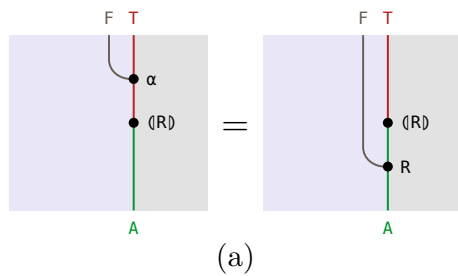
a relation  $X$  satisfying the fold equation  $\alpha X = F(X)R$  of  $R$  makes  $\frac{X}{\exists}$  the fold of the map  $\frac{F(\exists)R}{\exists}$



- (a)  $\alpha^\circ$  is a map because  $\alpha$  is an iso, so it leaves  $\Lambda$ :  $\frac{\alpha \circ F(X)R}{\exists} = \alpha \circ \frac{F(X)R}{\exists}$  — pow-laws  
 (b)  $F(\frac{X}{\exists}) = F(\frac{X}{\exists})F(\exists)$  as  $F$  is a functor, and  $F(\frac{X}{\exists})$  is a map (a relator sends maps to maps), so it leaves  $\Lambda$  —  $\triangleq$  relator, pow-laws  
 (c) fold uniqueness,  $\frac{F(\exists)R}{\exists}$  being a map — (11.4a)

$$\alpha \langle R \rangle = F(\langle R \rangle)R$$

the  $\Leftarrow$  half:  $\langle R \rangle$  satisfies the fold equation of  $R$



- (a) fold equation, cancellation — (11.6.2b), (11.4a), pow-laws,  $\triangleq$  relator

§11.6.3.  $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$ 

(11.6.3a)

$\varphi$  a **monotonic** mapping of the hom-set  $A \rightarrow B$  into itself:  $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$ .

$(\mu X : \varphi(X))$  the least  $X : A \rightarrow B$  with  $\varphi(X) \sqsubseteq X$ .

(11.6.3b)

| the law                                                                                    | what it says                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$                  | to bound $(\mu X : \varphi(X))$ above, exhibit one $Y$ the body does not grow past — the half §Section 11.6.4 and every chapter after it uses                                   |
| $\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$ | <b>Knaster–Tarski</b> : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$ , so the least prefix point and the least fixed point are one relation |

§11.6.4.  $(S)^\circ (R) = (\mu X : S^\circ F(X) R)$ 

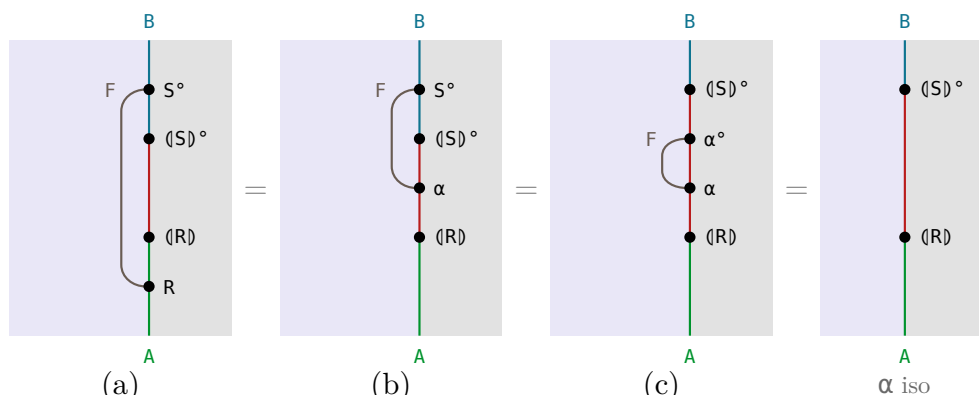
(11.6.4a)

$$(S)^\circ (R) = (\mu X : S^\circ F(X) R)$$

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$S^\circ F((S)^\circ (R)) R = (S)^\circ (R)$$

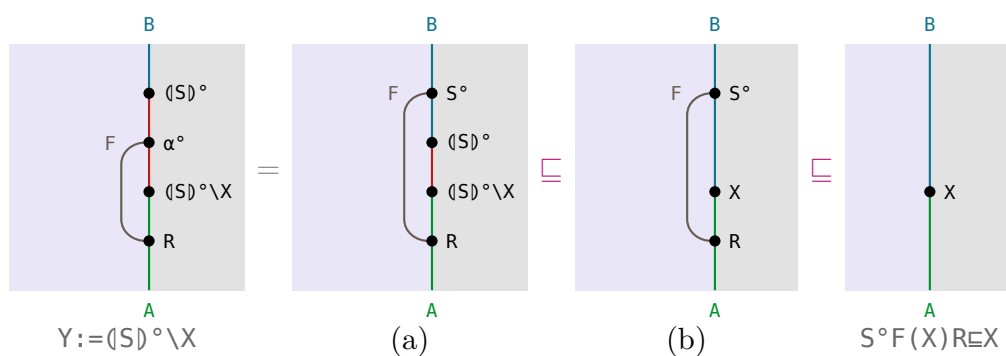
hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term  $S^\circ$  represents the decomposition stage,  $F((S)^\circ (R))$  the stage of solving the subproblems recursively, and  $R$  the recombination stage;  $R : FA \rightarrow A$ ,  $S : FB \rightarrow B$ ,  $\alpha : FT \rightarrow T$  initial



- (a) the body at  $(S)^\circ (R)$   
 $F(RS) = F(R)F(S) \triangleq \text{relator}$   
 (b)  $F((R))R = \alpha(R) \triangleq \text{reduce}$   
 (c)  $(S)^\circ \alpha^\circ = S^\circ F((S)^\circ) \triangleq \text{reduce, (11b)}$

$$S^\circ F(X) R \sqsubseteq X \Rightarrow (S)^\circ (R) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism  $(S)^\circ (R)$  is included in  $X$  if  $X$  satisfies the associated recursion inequation



- $Y := (S)^\circ \setminus X$   
 (a)  $(S)^\circ \alpha^\circ = S^\circ F((S)^\circ)$   
 $F(RS) = F(R)F(S) \triangleq \text{relator}$   
 (b)  $(S)^\circ ((S)^\circ \setminus X) \sqsubseteq X \triangleq \text{adj-all}$
- $S \cdot \dashv S \setminus \triangleq \text{adj-all}$
- (a)  $(R) = (\mu X : \alpha^\circ F(X) R) \triangleq \text{reduce, (11.6.3b), adj-all}$